

Math 2551 Worksheet: Lagrange Multipliers

1. A rectangular box without a lid is to be made from 12 m^2 of cardboard. Find the maximum volume of such a box.
2. Find the extreme values of the function $f(x, y) = x^2 + 2y^2$ on the circle $x^2 + y^2 = 1$.
3. Find the extreme values of the function $f(x, y, z) = x^2 + y^2 + z^2$ subject to the constraint $x^4 + y^4 + z^4 = 1$.
4. The plane $x + y + 2z = 2$ intersects the paraboloid $z = x^2 + y^2$ in an ellipse. Find the points on this ellipse that are nearest to and farthest from the origin. Hint: It may be helpful algebraically to work with the square of the distance to the origin.
5. Find the maximum volume of a rectangular box that is inscribed in a sphere of radius r .